

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Atmospheric pollution: smoke, acid rain, NO_x, hydrocarbons, CO, smogs - Set 1

Mark scheme

AQA spec 3.4.3.2.1-3.4.3.2.5; Pollution; textbook pp. 245-254

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) • Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect)	2
01.2	• $(930 - 250) / 250 \times 100$ • = 272%	2
01.3	• Anticyclonic conditions: clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled • Low wind velocity so air layers did not mix; London lies in a valley / basin where cold dense air collected: a temperature inversion formed • Smoke from coal fires was trapped below the inversion; moist air cooled to its dew point forming fog; smoke + fog = smog; the smog's high albedo prolonged the inversion	3
01.4	• Smoke particles (PM₁₀) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections • These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal	2
01.5	• Clean Air Act (1956)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
02.1	• Effluent gases pass through a chamber containing electrically charged wires / plates • Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick • Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials)	3

Question	Marking guidance	Marks
02.2	- Effluent gases are forced to rotate / spin in a cylindrical chamber - The (denser) particles are thrown outwards to the chamber walls where they fall and collect; cleaned gases leave through a central pipe	2
02.3	The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke	1
02.4	- Turbochargers deliver more air / oxygen to the combustion chamber - So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration)	2
02.5	- Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled - They penetrate deeper into the lungs / alveoli and are not cleared; may carry toxic / carcinogenic chemicals	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
03.1	- pH 5.6 - Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid	2
03.2	- Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) - Granite does not neutralise acids so lake water in A is acidic (4.6) - Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) - Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost	4
03.3	- Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines - Dissolves / weakens limestone and marble buildings and statues (eg York Minster)	2
03.4	- Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide - Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO_2 alone gives the weaker sulfurous acid)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	
04.1	- Most sulfur in coal is present as (solid) iron pyrites (FeS_2) - Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming)	2
04.2	- Flue gases are passed through a bed of crushed calcium carbonate (limestone) - Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) - If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster	3
04.3	- The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide - The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource	2

Question	Marking guidance	Marks
04.4	• Fluidised bed: increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NO_x (production prevented) • Catalytic converter: platinum / palladium catalyst chemically reduces NO_x back to nitrogen and oxygen gases in the exhaust • Urea sprayed into the hot flue / exhaust gases reacts with NO_x, converting it to nitrogen, carbon dioxide and water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	
05.1	• Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening • Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight • NO₂ (from morning traffic) is broken down by UV to NO and monatomic oxygen; monatomic oxygen reacts with O₂ to form ozone • Ozone lags behind the NO / NO₂ peaks (which are produced by the morning rush-hour traffic) because the reactions take time; falls at night when there is no sunlight	4
05.2	• NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure • NO is oxidised in the atmosphere (by oxygen / ozone) to NO₂, a secondary pollutant	2
05.3	• PANs (peroxyacetyl nitrates) • Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NO_x)	2
05.4	<i>Any 2 from:</i> • Heavy vehicle use giving high NO_x and hydrocarbon emissions • Sunny climate providing the UV for photochemical reactions • Valley / basin topography and frequent temperature inversions trapping pollutants	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
06.1	<i>Any 3 from:</i> • Unburnt fuel in vehicle exhaust gases • Evaporation from fuel tanks / spillage at filling stations • Solvents (paints, cleaning products) • Aerosol propellants • Gas leaks / fossil fuel exploitation (methane)	3
06.2	• Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon	2
06.3	• A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) • NO_x are chemically reduced to nitrogen and oxygen	3

Question	Marking guidance	Marks
06.4	• Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Smoke: solid particulates (PM10, PM5, PM1) from incomplete combustion; small particles stay suspended, penetrate lungs; toxic chemicals (acids, phenols, metals) carried on particles; high albedo cools climate; blocks light reducing photosynthesis; damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths); synergism with SO₂ • Controls: Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion; smoke smogs have largely been eliminated in UK cities • Acid rain gases: SO₂ (coal combustion, smelting sulfide ores) oxidised by ozone to SO₃ - sulfuric acid; NO_x (high temperature combustion) - nitric acid; HCl from PVC incineration; reactive gases travel long distances (transboundary pollution) • Effects: pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata); dissolves exoskeletons; leaches Ca and Mg then mobilises toxic Al and Pb; corrodes metals, dissolves limestone; severity depends on bedrock (limestone neutralises) • Controls: fuel desulfurisation (coal washing, hydrosulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays; switching from coal to gas / renewables • Evaluation: SO₂ emissions in Europe have fallen dramatically and lake pH is recovering; NO_x harder to control because of growing vehicle numbers; catalytic converters need lead-free fuel and warm engines; FGD is costly and produces wastes / gypsum; controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO₂ vs NO_x	9
07.2	<i>Any 1 from:</i> • Reduces growth rates of trees / crops • Damages leaf cuticles (allowing SO₂ to cause more damage)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	
08.1	• $(6400 - 220) / 6400 \times 100$ • = 96.6% (accept 97%)	2
08.2	<i>Any 3 from:</i> • Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO₂ source • Flue gas desulfurisation fitted to remaining coal plants; coal / oil desulfurisation • NO_x comes mainly from road transport and vehicle numbers / distance travelled increased greatly • NO_x is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur • Catalytic converters only work when warm / diesel engines produce more NO_x / older vehicles remained in use	3
08.3	• Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO₂ (broken down by UV to monatomic oxygen which forms ozone) • If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form	2
08.4	• Lichens are very sensitive to acidic conditions / SO₂ and different species have different tolerances • The species present, their abundance, size / health can be used as a biotic index of the pollution level • They give a long-term integrated measure of pollution rather than a single instantaneous reading; cheap, no equipment needed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	